FIITJEE

ALL INDIA TEST SERIES

FULL TEST – II

JEE (Main)-2020

TEST DATE: 22-12-2019

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

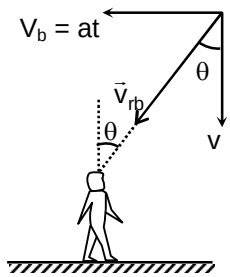
1. A

Sol. $\vec{v}_{r,b} = \vec{v}_r - \vec{v}_b$

$\tan\theta = \frac{at}{v} = t$

$\Rightarrow \sec^2\theta \frac{d\theta}{dt} = 1$

$\frac{d\theta}{dt} = \frac{1}{\sec^2\theta} = \frac{1}{(1+t^2)}$



2. C

Sol. Work done = area under P-t graph
(as t increases, area increases)

3. B

Sol. Using Mosely's law for both cobalt and impurity

$\sqrt{\nu} = K(Z-1)$

$\Rightarrow \sqrt{\frac{c}{\lambda}} = K(Z-1)$

$\Rightarrow \sqrt{\frac{c}{\lambda_{CO}}} = K(Z_{CO}-1)$ and $\sqrt{\frac{c}{\lambda_x}} = k(Z_x-1)$

$\sqrt{\frac{\lambda_{CO}}{\lambda_x}} = \frac{Z_x-1}{Z_{CO}-1} \Rightarrow Z_x = 40$

4. B

Sol. given that, perimeter is decreasing at a rate of 2v

$\therefore -\frac{d}{dt}(2\pi r) = 2v$

$\Rightarrow \frac{dr}{dt} = -\frac{v}{\pi}$

$\phi = B\pi r^2$

$\varepsilon = \left|-\frac{d\phi}{dt}\right|$

$\varepsilon = B2\pi r\left|\frac{dr}{dt}\right| = B2\pi r\frac{v}{\pi} = 2Brv$

$\Rightarrow I = \frac{\varepsilon}{R} = \frac{2Brv}{R}$

$I = \frac{2Brv}{\lambda(2\pi r)}$

$I = \frac{Bv}{\pi\lambda}$

5. C

Sol. $\omega = \sqrt{\frac{k}{m}}$

At equilibrium $k(\ell - a) = mg$

$\Rightarrow \quad \frac{k}{m} = \frac{g}{(\ell - a)}$

$\Rightarrow$ Angular frequency, $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{g}{(\ell - a)}}$

6. B

Sol. $\Delta r = (\mu_A - 1)t_A - (\mu_B - 1)t_B$

$= \mu_A t_A - t_A - \mu_B t_B + t_B$

$= t_B - t_A$

$\Rightarrow$ If $\Delta r > 0$, then fringe pattern will shift upwards (towards A)

$\Rightarrow$ If $\Delta r < 0$, then fringe pattern will shift downwards (towards B)

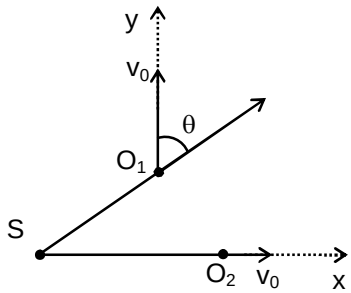
7. D

Sol. $f_1 = f_0\left(\frac{v - v_0\cos\theta}{v}\right)$

$f_2 = f_0\left(\frac{v - v_0}{v}\right)$

$f_1 > f_2 \qquad (\because v_0\cos\theta < v_0)$

$\Rightarrow \quad \left(\frac{f_1}{f_2}\right) > 1$



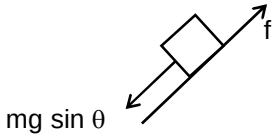
8. A

Sol. $mg\sin\theta = \eta A\frac{v}{t}$

$a^3\rho g\times\frac{3}{5} = \eta\times a^2\times\frac{v}{t}$

$\frac{3}{5}a\rho g = \frac{\eta v}{t}$

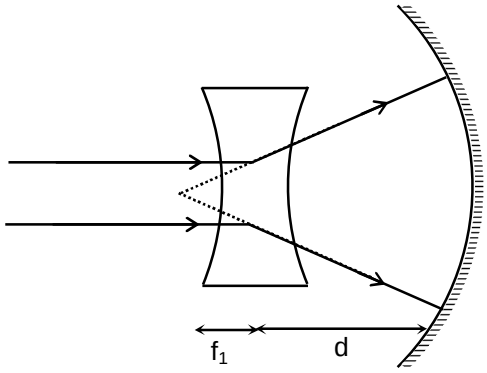
$\eta = \frac{3\rho agt}{5v}$



9. B

Sol. $(d + |f_1|) = 2|f_2|$

$\Rightarrow d = 2|f_2| - |f_1|$



10. B

Sol. As charge on the capacitor is conserved

$\Rightarrow C_0V_0 = C'V$

$\Rightarrow \frac{\varepsilon_0 A}{d_0}V_0 = \frac{\varepsilon_0 A}{d}V$

$\Rightarrow \frac{V_0}{d_0} = \frac{V_0}{d}\left(1+\frac{1}{2}\sin\omega t\right)$

After putting the values

$d_1 = \frac{d_0}{2}$

11. A

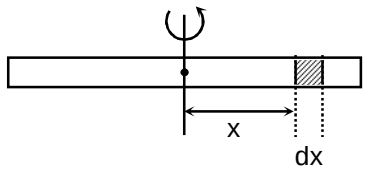
Sol. Current, $I = \left(\frac{V}{\rho d/S}\right) = \frac{VS}{\rho d}$

12. A

Sol. $dM = IA = (dq)f\pi x^2$

$= \frac{q}{\ell}dxf\pi x^2$

$M = \frac{qf\pi}{\ell}\int_{-\ell/2}^{\ell/2} x^2dx = \frac{q\pi f\ell^2}{12}$



13. D

Sol. $\vec{E}$ is parallel to $\vec{B}$. $\vec{v}_0$ is perpendicular to both. Therefore path of the particle is a helix with increasing pitch. Speed of particle at any time t is

$v = \sqrt{v_x^2 + v_y^2 + v_z^2}$ …(i)

Here, $v_y^2 + v_z^2 = v_0^2$

$v = 2v_0$

putting this in equation (i)

$v_x = \sqrt{3}v_0$ and $v_x = \frac{qE_0}{m}t$

$t = \frac{\sqrt{3}mv_0}{qE_0}$

14. A

Sol. $I = \frac{24}{\frac{RG}{R+G}+1}$

V = 24 – potential difference across 1 Ω

= 24 – I

$V = 24 - \left(\frac{24}{\frac{1}{\frac{I}{R}+\frac{1}{G}}+1} \right)$

For $G \to \infty \Rightarrow \frac{1}{G} \to 0$ and V = 20 volts

$\Rightarrow 20 = 24 - \frac{24}{R+1} \Rightarrow R = 5\Omega$

15. C

Sol. $A = \frac{\frac{d}{dt}\left[\int \vec{F}.d\vec{s}\right]}{\left[\vec{F}\cdot\vec{P}\right]}$

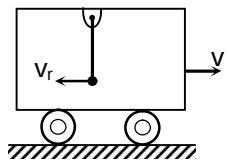
$= \frac{\left[\text{work}\right]}{\text{time} \times \frac{\text{work}}{\text{displacement}} \times P} = \frac{1}{M}$

16. A

Sol. Using conservation of momentum :

$m(v_r - v) = 6mv$

$\Rightarrow v_r = 7v$ …(i)



Now, using conservation of energy:

$mg\ell = \frac{1}{2}m(6v)^2 + \frac{1}{2}6mv^2$

$\Rightarrow v = \sqrt{\frac{g\ell}{21}}$

$\therefore v_r = 7v = \sqrt{\frac{7g\ell}{3}}$

17. C

Sol. After each collision normal components of velocity gets reversed.

$\therefore$ after 1st collision, velocity of coin, $\vec{v}_1 = 2\hat{i} - 3\hat{j}$

after 2st collision, velocity of coin, $\vec{v}_2 = -2\hat{i} - 3\hat{j}$

after 3rd collision, velocity of coin, $\vec{v}_3 = -2\hat{i} + 3\hat{j}$

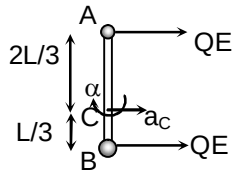
18. B

Sol. $E_1 = \dfrac{KP_1}{r^3}$

$F = P_2\left|\dfrac{dE_1}{dr}\right| = \dfrac{3KP_1P_2}{r^4}$

19. D

Sol. Acceleration of centre of mass of the system,



$a_C = \dfrac{2QE}{3m}$

$QE\left(\dfrac{2L}{3}\right) - QE\left(\dfrac{L}{3}\right) = \left(m \times \dfrac{4L^2}{9} + 2m \times \dfrac{L^2}{9}\right)\alpha$

$\dfrac{QEL}{3} = \dfrac{2mL^2}{3}\alpha$

$\alpha = \dfrac{QE}{2mL}$ clockwise

$a_A = a_C + \alpha\left(\dfrac{2L}{3}\right) = \dfrac{2QE}{3m} + \dfrac{QE}{2mL}\left(\dfrac{2L}{3}\right) = \dfrac{QE}{m}$

20. D

Sol. As it passes twice the transmission axis in one rotation.

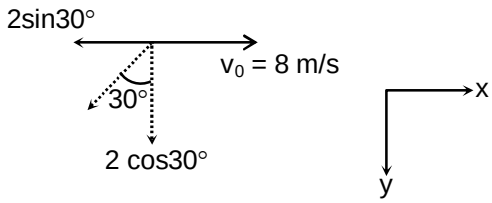
SECTION – B

21. 4

Sol. $v_x^2 + v_y^2 = 8^2$

$(8 - t)^2 + (\sqrt{3}t)^2 = 8^2$

$\Rightarrow t = 4$ sec



22. 1

Sol. Increase in length (dx) = $\alpha(dx)\Delta T$

$(d\ell) = = \dfrac{x}{5000} \times (dx) \times 100 = \dfrac{x}{50}dx$

$\therefore$ total expansion, $\Delta\ell = \displaystyle\int_0^1 \frac{x}{50}\,dx = \frac{1}{100}m = 1\,cm$.

SECTION – C

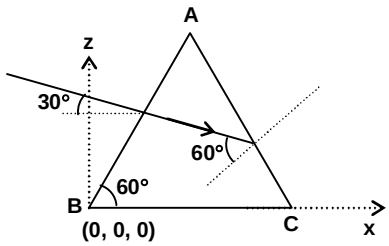
23. 00001.15

Sol. Since, $\sqrt{3}z + x = 10$

$\Rightarrow z = -\frac{x}{\sqrt{3}} + \frac{10}{\sqrt{3}}$

Slope = $-\frac{1}{\sqrt{3}} = \tan\theta$

$\Rightarrow \theta = -30°$



So, the ray is incident normally on the face AB. Thus angle of incidence on the face AC is 60° for grazing the face AC,

$\mu \sin(60°) = 1 \times \sin 90°$

$\mu = \frac{2}{\sqrt{3}} = 1.1547$

24. 00010.00

Sol. According to the problem, power = 1W.

Zener breakdown voltage, $V_z = 5V$

Minimum voltage, $V_{min} = 3V$

Maximum voltage, $V_{max} = 7V$

We know, P = VI

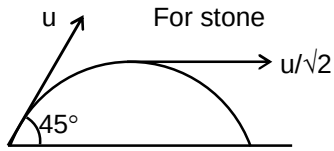
So, current $I_{Z_{max}} = \frac{P}{V_Z} = \frac{1}{5} = 0.2\,A$

For safe operation, $R_s = \frac{V_{max} - V_z}{I_{Z_{max}}} = \frac{7-5}{0.2} = 10\,\Omega$

25. 00019.60

Sol. For stone, at the highest point

Radius of curvature, $R = \frac{\left(u/\sqrt{2}\right)^2}{g} = \frac{u^2}{2g}$



The bee follows the stone's trajectory with the constant speed u

$a = \frac{u^2}{R} = \frac{u^2}{u^2/2g} = 2g$

$a = 2g = 19.6\ m/s^2$

Chemistry

PART – II

SECTION – A

26. C

Sol. For adiabatic process

$$\frac{T_2}{T_1}=\left(\frac{V_1}{V_2}\right)^{\gamma-1}=\left(\frac{P_1}{P_2}\right)^{\frac{1-\gamma}{\gamma}}$$

$$\Rightarrow \frac{T_2}{T_1}=\left(\frac{V_1}{V_2}\right)^{\frac{R}{C_V}}=\left(\frac{P_2}{P_1}\right)^{\frac{R}{C_P}}$$

$$\text{or } \ln\frac{T_2}{T_1}=\frac{R}{C_V}\ln\frac{V_1}{V_2}=\frac{R}{C_P}\ln\frac{P_2}{P_1}$$

27. D

Sol. A $\rightarrow$ Aromatic

B $\rightarrow$ Non-aromatic

C $\rightarrow$ Aromatic

D $\rightarrow$ Antiaromatic

28. D

Sol. Sucrose is non-reducing disaccharide.

29. D

Sol. Due to hydrogen bonding.

30. A

Sol. pH = 6 $\therefore H^+ = 10^{-6}$

$$HIn \rightleftharpoons H^+ + In^-$$

C	0	0
$\frac{C}{2}$	$\frac{C}{2}$	$\frac{C}{2}$

$$K_{HIn}=\frac{\left[H^+\right]\left[In^-\right]}{\left[HIn\right]}$$

$$=\frac{10^{-6}\times C/2}{C/2}=10^{-6}$$

31. C

Sol. Correct order for bond dissociation energy is

	Cl_2	>	Br_2	>	F_2	>	I_2
kJ/mol	(242)		(193)		(159)		(151)

32. B

Sol. Number of X atom left in unit cell $= 4-\frac{1}{8}\times 5-\frac{1}{2}\times 1$

$= \frac{32-5-4}{8} = \frac{23}{8}$

Number of Y atom left in unit cell $= 4 - \frac{1}{4} \times 4 - 1 = 2$

Number of Z atom left in unit cell $= 8 - 2 = 6$

$\therefore$ Formula = $X_{\frac{23}{8}}Y_2Z_6$

= $X_{23}Y_{16}Z_{48}$

33. D

34. A

Sol. SO_2 is a non-linear molecule.

$\therefore$ Translational degree of freedom = 3

Rotational degree of freedom = 3

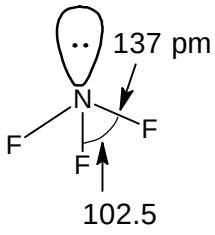
Vibrational degree of freedom = (3 × 3 – 6) = 3

But, vibration degree of freedom is inactive at low temperature.

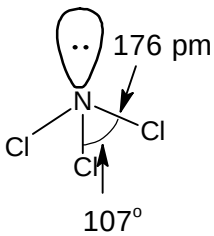
$\therefore$ Active degree of freedom = 3 + 3 = 6

35. B

Sol.



102.5



107°

$\mu = 0.24$ D	0.37 D
M.P. = 66 K	230 K

36. C

37. B

Sol. 'A' and 'C' has place of symmetry. 'D' is a planar molecule.

38. A

Sol. $PV = \frac{W}{M}RT$

$\Rightarrow \frac{T}{M} = \frac{PV}{WR} = \frac{5 \times 1.5}{10 \times 0.0821} = 9.135$ K / gm = 9.135×10^3 K/kg

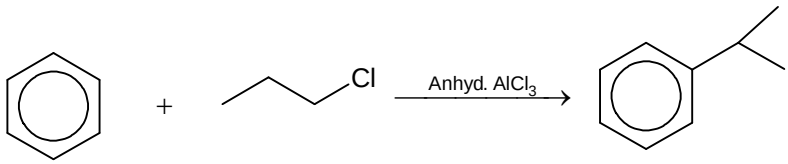
$V_{rms} = \sqrt{3R\left(\frac{T}{M}\right)} = \sqrt{3 \times 8.314 \times 9.135 \times 10^3}$

= 477.33 m/sec.

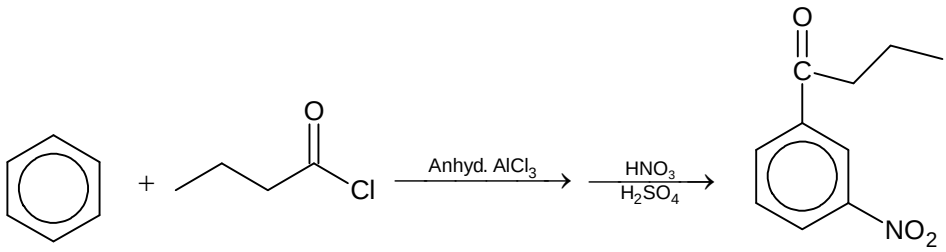
39. A

40. C

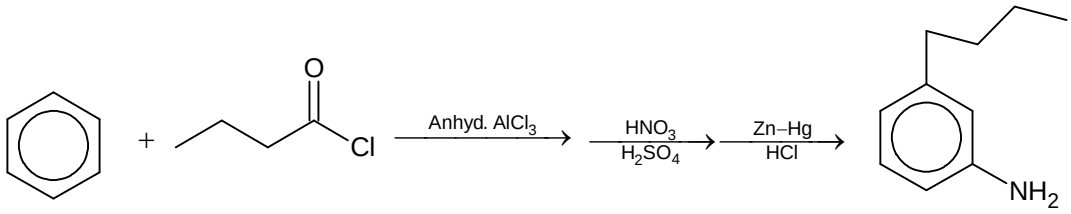
Sol. (A)



(B)

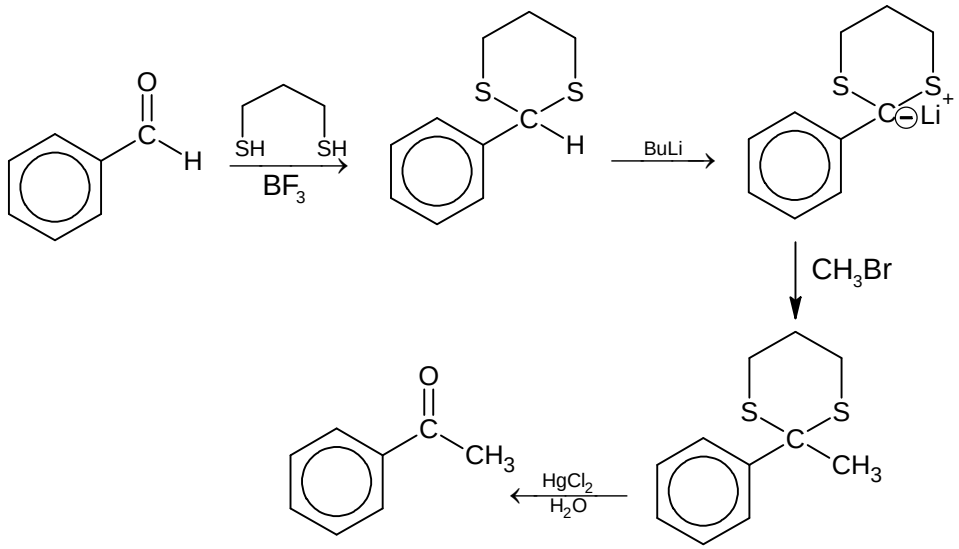


(D)



41. D

Sol.



42. D

Sol. Order of enol content in pure liquid state
D > C > B > A

43. B

44. D

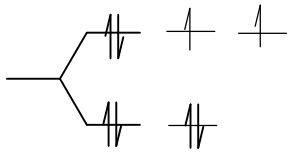
Sol. In PbO_2 Pb is in +4 oxidation state.

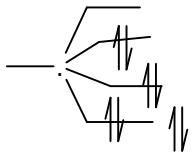
45. D

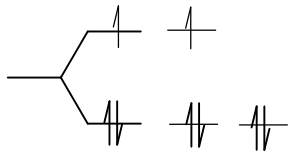
SECTION – B

46. 4

Sol.

$[NiCl_4]^{2-}$ Ni^{2+} (Td)  $n_1 = 2$

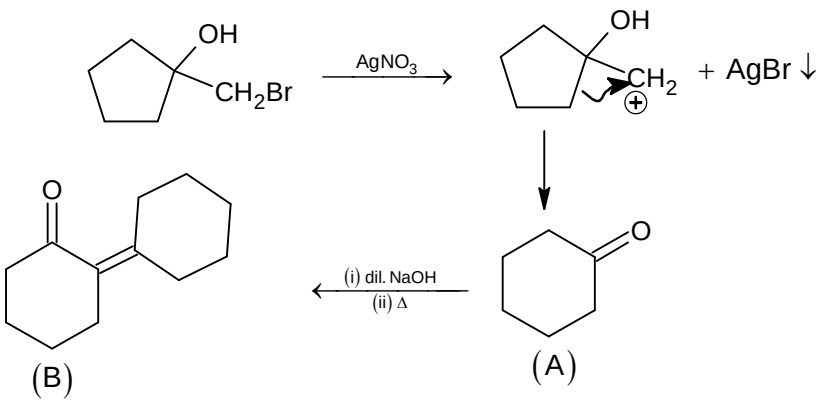
$[Ni(CN)_4]^{2-}$ Ni^{2+} (Sq. planar)  $n_2 = 0$

$[Ni(H_2O)_6]^{2+}$ Ni^{2+} (Oh)  $n_3 = 2$

Total number of unpaired electrons = $n_1 + n_2 + n_3 = 4$

47. 4

Sol.



Degree of unsaturation = 4.

SECTION – C

48. 00000.40

Sol. $K_P = \dfrac{\left(P_{SO_3}\right)^2}{\left(P_{SO_2}\right)^2\left(P_{O_2}\right)}$

$10 = \dfrac{\left(P_{SO_3}\right)^2}{\left(\dfrac{P_{SO_3}}{2}\right)^2 . P_{O_2}}$

$10 = \dfrac{4}{P_{O_2}}$

$\therefore P_{O_2} = \dfrac{4}{10} = 0.4 \text{ atm}$

49. 00003.25

Sol. $K_3\left[Fe(CN)_6\right] \longrightarrow 3K^+ + \left[Fe(CN)_6\right]^{3-}$

C	0	0
$C - C\alpha$	$3C\alpha$	$C\alpha$

$$i = \frac{C - C\alpha + 3C\alpha + C\alpha}{C} = 1 + 3\alpha$$

Given $\alpha = 0.75$

$\therefore i = 1 + 3 \times 0.75 = 3.25$

50. 00004.25

Sol. M.eq. of NH_3 = 500 × 0.6 – 500 × 0.1
= 250 M.eq.

$$\frac{W}{E} \times 1000 = 250$$

$$\therefore W = \frac{250 \times 17}{1000} = 4.25\,gm$$

Mathematics

PART – III

SECTION – A

51. C

Sol. Centre of hyperbola is (3, 4) one end of transverse axis is (3, –1)

$\therefore$ Transverse axis is parallel to y-axis $\Rightarrow$ b = 5 and $b^2 - a^2 = 20 \Rightarrow a = \sqrt{5}$

$\therefore$ Equation of hyperbola is $\frac{(x-3)^2}{5} - \frac{(y-4)^2}{25} = -1$

52. C

Sol. $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$ are in A.P., $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \beta\gamma + \gamma\alpha}{\alpha\beta\gamma} = \frac{-54/10}{27/10} - 2 \Rightarrow \frac{3}{\beta} = -2 \Rightarrow \beta = \frac{-3}{2}$

$\therefore 10\left(\frac{-27}{8}\right) - m\left(\frac{9}{4}\right) - 54\left(\frac{-3}{2}\right) - 27 = 0 \Rightarrow m = 9$

53. A

Sol. Last digit of $(2017)^{2018}$ is 9, last digit of $(2018)^{2019}$ is 2, last digit of $(2019)^{2020}$ is 1

54. C

Sol. Add and subtract 7! , $7! + 8! + \frac{9!}{2!} + \frac{10!}{3!} + ... \frac{27!}{20!} - 7! = 7!\left(1 + \frac{8!}{1!7!} + \frac{9!}{2!7!} + \frac{10!}{3!7!} + ... \frac{27!}{20!7!}\right) - 7!$

$7!\left({}^7C_0 + 8C_1 + 9C_2 + 10C_3 + ... {}^{27}C_{20}\right) - 7! = 7!\left(8C_0 + 8C_1 + 9C_2 + 10C_3 + ... {}^{27}C_{20}\right) - 7!$

$= 7!\left({}^{28}C_{20}\right) - 7!$

55. D

Sol. $C^{-1}B = AC^{-1} \Rightarrow C^{-1}BC = A \Rightarrow A^2 = C^{-1}BCC^{-1}BC = C^{-1}B^2C$

$\Rightarrow A^3 = C^{-1}B^2CC^{-1}BC = C^{-1}B^3C$

Similarly proceeding $A^5 = C^{-1}B^5C \therefore CA^5C^{-1} = CC^{-1}B^5CC^{-1} = B^5$

56. B

Sol. $\begin{vmatrix}\alpha_1 & \alpha_2 \\ \beta_1 & \beta_2\end{vmatrix} = 0 \Rightarrow \alpha_1\beta_2 - \alpha_2\beta_1 = 0 \Rightarrow \frac{\alpha_1}{\beta_1} = \frac{\alpha_2}{\beta_2}$

Using componendo and dividendo $\frac{\alpha_1 + \beta_1}{\alpha_1 - \beta_1} = \frac{\alpha_2 + \beta_2}{\alpha_2 - \beta_2} \Rightarrow \frac{-b/a}{\sqrt{\frac{b^2}{a^2} - \frac{4c}{a}}} = \frac{-q/p}{\sqrt{\frac{q^2}{p^2} - \frac{4r}{p}}}$

$\Rightarrow \frac{b}{\sqrt{b^2 - 4ac}} = \frac{q}{\sqrt{q^2 - 4pr}} \Rightarrow b^2pr = q^2ac$

57. D

Sol. $|z|^2 = 9 \Rightarrow z\bar{z} = 9 \Rightarrow \left|3 + \frac{9}{z}\right|^2 + |3 - z|^2 \Rightarrow \left|3 + \frac{9}{9/\bar{z}}\right|^2 + |3 - z|^2$

$\Rightarrow |3 + \bar{z}|^2 + |3 - z|^2 = 2\left(9 + |z|^2\right) = 36$

58. B

Sol. Let point P be (h, k) $\therefore$ Equation of pair of tangents is $T^2 = SS_1$

$\Rightarrow (ky - 4x - 4h)^2 = (y^2 - 8x)(k^2 - 8h) \Rightarrow 16x^2 + y^2(8h) - 8xyk - 32hx + 8xk^2 - 16h^2 + 8hky = 0$

$\because$ Angle is $45° \Rightarrow \tan 45° = 1 = \dfrac{2\sqrt{16k^2 - 16 \times 8h}}{16 + 8h} \Rightarrow 16 + 8h = 2 \times 4\sqrt{k^2 - 8h}$

$\Rightarrow 2 + h = \sqrt{k^2 - 8h} \Rightarrow 4 + h^2 + 4h = k^2 - 8h \Rightarrow h^2 + 12h - k^2 + 4 = 0 \Rightarrow (x + 6)^2 - y^2 = 32$

59. C

Sol. $\dfrac{x\,dy - y\,dx}{x^2} = \dfrac{x\sqrt{1 - \left(\frac{y}{x}\right)^2}\,dx}{x^2} \Rightarrow d\left(\dfrac{y}{x}\right) = \dfrac{\sqrt{1 - \left(\frac{y}{x}\right)^2}\,dx}{x} \Rightarrow \dfrac{d(y/x)}{\sqrt{1 - \left(\frac{y}{x}\right)^2}} = \dfrac{dx}{x}$

$\Rightarrow \sin^{-1}\left(\dfrac{y}{x}\right) = \ln x + c$

At $x = 1, y = 0 \Rightarrow 0 = 0 + c \Rightarrow c = 0$

$\Rightarrow \sin^{-1}\left(\dfrac{y}{x}\right) = \ln x$

$\Rightarrow y = x \sin(\ln x)$

60. B

Sol. Equation of AB is $x^2 + 4y^2 + \alpha xy = 0$ ..... (1)

AB is chord of contact with respect to point P

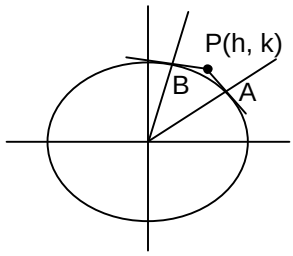
$\Rightarrow$ Equation of AB is $hx + 4ky = 4$

Combined equation of AB is $x^2 + 4y^2 - 4\left(\dfrac{hx + 4ky}{4}\right)^2 = 0$

$\Rightarrow x^2(4 - h^2) + y^2)(16 - 16k^2) - 8hxy = 0$ ..... (2)

Equation (1) and (2), represent same equation

$\therefore \dfrac{4 - h^2}{1} = \dfrac{16 - 16k^2}{4} = \dfrac{-8hk}{\alpha} \Rightarrow x^2 = 4y^2$



61. B

Sol. Let equation of circle is $x^2 + y^2 + 2gx + 2fy + c = 0 \Rightarrow 1 + 2f + c = 0$

$2 \times g \times (-1) + 2 \times f \times 0 = -15 + c$ and $2 \times g \times 0 + 2f \times 0 = -1 + c \Rightarrow c = 1$

$-2g = -15 + c$

$\Rightarrow g = 7 \Rightarrow f = -1$

62. B

Sol. Let $y = mx + \sqrt{25m^2 + 4}$ be a tangent to a given ellipse

$\because$ It is a common tangent $\therefore \dfrac{\left|0 - 0 + \sqrt{25m^2 + 4}\right|}{\sqrt{m^2 + 1}} = 4 \Rightarrow m = \pm\dfrac{2}{\sqrt{3}}$

$\therefore$ Equation of common tangent is $y = \pm\dfrac{2}{\sqrt{3}}x \pm \dfrac{\sqrt{112}}{\sqrt{3}}$

$\therefore$ Length of intercept = $\dfrac{14}{\sqrt{3}}$

63. C

Sol. E_k : k white balls and (5 – k) black balls are taken from bag A

C : bag C contains 3 white balls exactly

$$P(C) = P\left(\frac{C}{E_3}\right)P(E_3) + P\left(\frac{C}{E_4}\right)P(E_4) + P\left(\frac{C}{E_5}\right)P(E_5)$$

$$= \frac{{}^5C_3 \times {}^5C_2}{{}^{10}C_5} \times \left(\frac{{}^3C_3 \times {}^2C_1}{{}^5C_4}\right) + \frac{{}^5C_4 \times {}^5C_1}{{}^{10}C_5} \times \left(\frac{{}^4C_3 \times {}^1C_1}{{}^5C_4}\right) + \frac{{}^5C_5}{{}^{10}C_5}\left(\frac{0}{{}^5C_4}\right)$$

$$= \frac{10 \times 10 \times 2}{{}^{10}C_5 \times 5} \times \frac{5 \times 5}{{}^{10}C_5} \times \frac{4}{5} + 0 = \frac{60}{{}^{10}C_5}$$

64. A

Sol. $\cos A + 2\cos B = 2 - 2\cos C \Rightarrow 2\cos\left(\frac{A+C}{2}\right)\cos\left(\frac{A-C}{2}\right) = 4\sin^2\frac{B}{2}$

$$\Rightarrow \cos\left(\frac{A-C}{2}\right) = 2\sin\frac{B}{2} \Rightarrow \cos\frac{A}{2}\cos\frac{C}{2} + \sin\frac{A}{2}\sin\frac{C}{2} = 2\sin\frac{B}{2}$$

$$\Rightarrow \sqrt{\frac{s(s-a)}{bc}}\sqrt{\frac{s(s-c)}{ab}} + \sqrt{\frac{(s-b)(s-c)}{bc}} \times \sqrt{\frac{(s-a)(s-b)}{ab}} = 2\sqrt{\frac{(s-a)(s-c)}{ac}}$$

$$\Rightarrow \frac{s}{b\sqrt{ac}} + \frac{s-b}{b\sqrt{ac}} = \frac{2}{\sqrt{ac}} \Rightarrow a + c = 2b$$

65. A

Sol. Standard deviation doesn't change if we increase or decrease each number by a particular amount

66. B

Sol. $\sim(p \wedge \sim q) \rightarrow \sim r$

$= \sim p \vee q \rightarrow \sim r$

67. C

Sol. $f'(x) = 0$ must have two real and unequal roots

$f'(x) = 3x^2 - 6(2\lambda - 1)x + 6\lambda = 0$ must have $D > 0 \Rightarrow 36(2\lambda - 1)^2 - 72\lambda > 0$

$\Rightarrow 4\lambda^2 + 1 - 4\lambda - 2\lambda > 0 \Rightarrow 4\lambda^2 - 6\lambda + 1 > 0,\ \lambda \in \left(-\infty, 3 - \sqrt{5}\right) \cup \left(3 + \sqrt{5}, \infty\right)$

68. C

Sol. Line makes equal angles with both planes and hence equal angles with normals of both planes

$$\frac{\lambda + 4 + 36}{\sqrt{\lambda^2 + 4 + 144}\ \sqrt{1^2 + 2^2 + 3^2}} = \frac{2\lambda + 6 + 12}{\sqrt{\lambda^2 + 4 + 144}\ \sqrt{1^2 + 2^2 + 3^2}} \ \therefore \lambda = 22$$

69. B

Sol. Let $g(x) = p(x)(3 + x) - 2$ ..... (1)

Then $g(x)$ is 4 degree polynomial whose roots are 0, 1, 2, 3

$$\Rightarrow g(x) = kx(x-1)(x-2)(x-3) \Rightarrow g(-3) = -2 \Rightarrow -2 = k \times -3 \times -4 \times -5 \times -6 \Rightarrow k = \frac{-1}{180}$$

$$\therefore\ g(x) = \frac{-x(x-1)(x-2)(x-3)}{180} = p(x)(3+x) - 2\text{, put } x = 6$$

$$\frac{-6 \times 5 \times 4 \times 3}{180} = p(6)9 - 2 \Rightarrow p(6) = 0$$

70. C

Sol. Integrating both sides $f(x) = \frac{2f(6-2x)}{-2} + c$

$f(x) + f(6 - 2x) = c$, put $x = 2$, $f(2) + f(2) = c \Rightarrow c = 8$
$\Rightarrow f(x) + f(6 - 2x) = 8$
Put $x = 1 \Rightarrow f(1) + f(4) = 8$

SECTION – B

71. 7

Sol. Divide numerator and denominator by $n5^n \Rightarrow \lim\limits_{n\to\infty} \dfrac{1}{\left(\dfrac{x-3}{5}\right)^n + 25 - \dfrac{1}{n}} = \dfrac{1}{25}$

$\Rightarrow -1 < \dfrac{x-3}{5} < 1 \Rightarrow -2 < x < 8$

72. 5

Sol. $\dfrac{\alpha^{16} - \beta^{16} - 6\left(\alpha^{14} - \beta^{14}\right)}{2\left(\alpha^{15} - \beta^{15}\right)} = \dfrac{\alpha^{14}\left(\alpha^2 - 6\right) - \beta^{14}\left(\beta^2 - 6\right)}{2\left(\alpha^{15} - \beta^{15}\right)} = \dfrac{10\alpha^{15} - 10\beta^{15}}{2\left(\alpha^{15} - \beta^{15}\right)} = 5$

SECTION – C

73. 00007.00

Sol. $I = \int \dfrac{x\left(1 - \dfrac{1}{x^3}\right)^{\frac{1}{4}}}{x^5}\,dx = \int \dfrac{\left(1 - \dfrac{1}{x^3}\right)^{\frac{1}{4}}}{x^4}\,dx$. Let $1 - \dfrac{1}{x^3} = t^4 \Rightarrow \dfrac{3}{x^4}dx = 4t^3dt$

$\Rightarrow I = \int t \times \dfrac{4}{3}t^3 dt = \dfrac{4}{15}\left(1 - \dfrac{1}{x^3}\right)^{\frac{5}{4}} + c$ $\therefore$ m = 4, k = 15

74. 00002.50

Sol. Foot of perpendicular of focus on any tangent lies on auxiliary circle

Let image be (h k), focus is (4, 0) or (–4, 0) $\Rightarrow \dfrac{h+4}{2}, \dfrac{k}{2}$ lies on auxiliary circle

$\therefore$ Locus is $(x + 4)^2 + y^2 = 100$

75. 00000.50

Sol. For equation of OA
$x + 4y - 3 + \lambda(ax - y + 3) = 0$
It passes through (0, 0) $\therefore \lambda = 1$
$\therefore$ Equation of OA is $(a + 1)x + 3y = 0$
Similarly equation of OB is $-2x + (4 - 3b)y = 0$
$\therefore \dfrac{-(a+1)}{3} \times \dfrac{2}{(4-3b)} = -1 \Rightarrow 2a + 2 = 12 - ab$
$\Rightarrow 2a + 9b = 10$

